

Lab 3 Fork and Exec

Make:

```
@stjamesoto →/workspaces/OS-Lab-Processes-and-Fork (main) $ make
gcc -Wall -Wextra -std=gnu11 -g challenges/ch1_single_fork.c -o bin/ch1_single_fork
gcc -Wall -Wextra -std=gnu11 -g challenges/ch2_multi_child.c -o bin/ch2_multi_child
gcc -Wall -Wextra -std=gnu11 -g challenges/ch3_exec_ls.c -o bin/ch3_exec_ls
gcc -Wall -Wextra -std=gnu11 -g challenges/ch4_exec_worker.c -o bin/ch4_exec_worker
gcc -Wall -Wextra -std=gnu11 -g challenges/ch5_exec_examples.c -o bin/ch5_exec_examples
gcc -Wall -Wextra -std=gnu11 -g challenges/ch6_pipe_sum.c -o bin/ch6_pipe_sum
gcc -Wall -Wextra -std=gnu11 -g challenges/ch7_pipeline.c -o bin/ch7_pipeline
gcc -Wall -Wextra -std=gnu11 -g challenges/ch8_wait_nonblock.c -o bin/ch8_wait_nonblock
gcc -Wall -Wextra -std=gnu11 -g challenges/ch9_zombie.c -o bin/ch9_zombie
gcc -Wall -Wextra -std=gnu11 -g challenges/worker.c -o bin/worker
```

Demo Screenshots:

3.1

```
@stjamesoto →/workspaces/OS-Lab-Processes-and-Fork (main) $ ./bin/fork_demo
[parent] created child PID=15152
[child] PID=15152, PPID=15151
[child] exiting with code 42
[parent] child 15152 exited with status 42
@stjamesoto →/workspaces/OS-Lab-Processes-and-Fork (main) $
```

3.2

```
@stjamesoto →/workspaces/OS-Lab-Processes-and-Fork (main) $ ./bin/exec_demo
[child] about to exec ls -l
total 32
-rw-rw-rw- 1 codespace root      1052 Aug 28 17:22 CMakeLists.txt
-rw-rw-rw- 1 codespace root       483 Aug 28 17:22 Makefile
-rw-rw-rw- 1 codespace codespace 4272 Aug 28 17:31 README.md
drwxrwxrwx+ 2 codespace codespace 4096 Aug 29 15:47 bin
drwxrwxrwx+ 2 codespace root      4096 Aug 28 17:22 challenges
drwxrwxrwx+ 2 codespace codespace 4096 Aug 29 15:42 lab_report
drwxrwxrwx+ 2 codespace root      4096 Aug 28 17:22 src
[parent] child finished exec
```

3.3

```
@stjamesoto →/workspaces/OS-Lab-Processes-and-Fork (main) $ ./bin/pipe_demo
5
```

3.4

```
@stjamesoto →/workspaces/OS-Lab-Processes-and-Fork (main) $ (cd src && ../bin/pipeline_demo)
exec_demo.c
fork_demo.c
pipe_demo.c
pipeline_demo.c
zombie_demo.c
```

3.5

```

@stjamesoto →/workspaces/OS-Lab-Processes-and-Fork (main) $ ./bin/zombie_demo & sleep 3; ps -l
[1]+  Done                  ./bin/ch9_zombie
[1] 33538
[parent] PID=33538 created child 33540; sleeping 10s (child is a zombie)
[parent] run `ps -l` in another terminal to see child in Z state
[child] PID=33540 exiting immediately
F S  UID    PID    PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
4 S  1000    2091    1133  0  80   0 -  4153 do_wai pts/1        00:00:00 bash
0 S  1000    33538    2091  0  80   0 -   671 hrtime pts/1        00:00:00 zombie_demo
1 Z  1000    33540    33538  0  80   0 -     0 -      pts/1        00:00:00 zombie_demo
0 R  1000    33598    2091  0  80   0 -   2872 -      pts/1        00:00:00 ps
@stjamesoto →/workspaces/OS-Lab-Processes-and-Fork (main) $ [parent] reaped child; zombie gone

1 ?      Ss      0:00 /sbin/docker-init -- /bin/sh -c echo Container started trap "exit 0" 15 /usr/local/shar
@stjamesoto →/workspaces/OS-Lab-Processes-and-Fork (main) $ ps -l
F S  UID    PID    PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
4 S  1000    25612    1133  0  80   0 -  4024 do_wai pts/0        00:00:00 bash
0 R  1000    35784    25612  0  80   0 -   2839 -      pts/0        00:00:00 ps

```

Coding Challenges:

1. Single Fork

```

@stjamesoto →/workspaces/OS-Lab-Processes-and-Fork (main) $ ./bin/ch1_single_fork
Hello from child (PID=20875, PPID=20874)
Parent: child 20875 exited with status 7

```

2. Multiple Children in a loop

```

Parent: child 20875 exited with status 7
@stjamesoto →/workspaces/OS-Lab-Processes-and-Fork (main) $ ./bin/ch2_multi_child 5
[child 0] PID=23243 exiting with code 1
[parent] (#0 finished) child 23243 exited with status 1
[child 4] PID=23247 exiting with code 5
[parent] (#1 finished) child 23247 exited with status 5
[child 1] PID=23244 exiting with code 2
[child 2] PID=23245 exiting with code 3
[parent] (#2 finished) child 23245 exited with status 3
[child 3] PID=23246 exiting with code 4
[parent] (#3 finished) child 23246 exited with status 4
[parent] (#4 finished) child 23244 exited with status 2
@stjamesoto →/workspaces/OS-Lab-Processes-and-Fork (main) $

```

3. execlp()

```
@stjamesoto →/workspaces/OS-Lab-Processes-and-Fork (main) $ (cd challenges && ../bin/ch3_exec_ls)
total 52
drwxrwxrwx+ 2 codespace root 4096 Aug 28 17:22 .
drwxrwxrwx+ 7 codespace root 4096 Aug 29 15:47 ..
-rw-rw-rw- 1 codespace root 2136 Aug 28 17:22 ch10_pool.c
-rw-rw-rw- 1 codespace root 698 Aug 28 17:22 ch1_single_fork.c
-rw-rw-rw- 1 codespace root 1136 Aug 28 17:22 ch2_multi_child.c
-rw-rw-rw- 1 codespace root 454 Aug 28 17:22 ch3_exec_ls.c
-rw-rw-rw- 1 codespace root 814 Aug 28 17:22 ch4_exec_worker.c
-rw-rw-rw- 1 codespace root 702 Aug 28 17:22 ch5_exec_examples.c
-rw-rw-rw- 1 codespace root 896 Aug 28 17:22 ch6_pipe_sum.c
-rw-rw-rw- 1 codespace root 1105 Aug 28 17:22 ch7_pipeline.c
-rw-rw-rw- 1 codespace root 1111 Aug 28 17:22 ch8_wait_nonblock.c
-rw-rw-rw- 1 codespace root 808 Aug 28 17:22 ch9_zombie.c
-rw-rw-rw- 1 codespace root 344 Aug 28 17:22 worker.c
[parent] child finished listing directory
```

4. Exec an internal helper program

```
@stjamesoto →/workspaces/OS-Lab-Processes-and-Fork (main) $ (cd bin && ../ch4_exec_worker)
worker: argv[0]=worker
worker: argv[1]=alpha
worker: argv[2]=beta
worker: MYVAR=hello
[parent] worker finished
```

5. Show execv vs execl

```
@stjamesoto →/workspaces/OS-Lab-Processes-and-Fork (main) $ ../bin/ch5_exec_examples
one two
one two
```

6. Parent-to-child pipe

```
@stjamesoto →/workspaces/OS-Lab-Processes-and-Fork (main) $ ../bin/ch6_pipe_sum
Sum = 55
```

7. Build a pipeline

```
@stjamesoto →/workspaces/OS-Lab-Processes-and-Fork (main) $ (cd challenges && ../bin/ch7_pipeline "\.c$")
ch10_pool.c
ch1_single_fork.c
ch2_multi_child.c
ch3_exec_ls.c
ch4_exec_worker.c
ch5_exec_examples.c
ch6_pipe_sum.c
ch7_pipeline.c
ch8_wait_nonblock.c
ch9_zombie.c
worker.c
```

8. waitpid()

```

• @stjamesoto → /workspaces/OS-Lab-Processes-and-Fork (main) $ ./bin/ch8_wait_nonblock
[parent] child 25342 finished (status 1)
[parent] child 25343 finished (status 2)
[parent] child 25344 finished (status 3)
[parent] all children reaped

```

9. Create and observe a zombie process

```

• @stjamesoto → /workspaces/OS-Lab-Processes-and-Fork (main) $ ./bin/ch9_zombie & sleep 3; ps -l
[1] 30337
[parent] PID=30337, child 30339 is now a ZOMBIE for 10s
[parent] observe with: ps -l (look for state Z)
[child] PID=30339 exiting now
F S  UID      PID      PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
4 S  1000      2091      1133  0  80   0 -  4024 do_wai pts/1      00:00:00 bash
0 S  1000      30337      2091  0  80   0 -   671 hrtime pts/1      00:00:00 ch9_z
1 Z  1000      30339      30337  0  80   0 -    0 -      pts/1      00:00:00 ch9_z
0 R  1000      30395      2091  0  80   0 -  2872 -      pts/1      00:00:00 ps
@stjamesoto → /workspaces/OS-Lab-Processes-and-Fork (main) $ [parent] child reaped; zombie removed

```

```

• @stjamesoto → /workspaces/OS-Lab-Processes-and-Fork (main) $ ps -l
F S  UID      PID      PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
4 S  1000      25612      1133  0  80   0 -  3991 do_wai pts/0      00:00:00 bash
0 R  1000      30575      25612  0  80   0 -  2839 -      pts/0      00:00:00 ps

```

10. Parallel worker pool

```

• @stjamesoto → /workspaces/OS-Lab-Processes-and-Fork (main) $ ./bin/ch10_pool 3 file1 file2 file3 file4 file5
[parent] started worker PID=27218 for 'file1' (active=1)
[parent] started worker PID=27219 for 'file2' (active=2)
  [worker PID=27219] processing 'file2'
[parent] started worker PID=27220 for 'file3' (active=3)
  [worker PID=27220] processing 'file3'
  [worker PID=27218] processing 'file1'
  [worker PID=27219] done 'file2'
  [worker PID=27220] done 'file3'
  [worker PID=27218] done 'file1'
[parent] worker PID=27218 finished (status 0, active=2)
[parent] started worker PID=27232 for 'file4' (active=3)
[parent] worker PID=27219 finished (status 0, active=2)
  [worker PID=27232] processing 'file4'
  [worker PID=27233] processing 'file5'
[parent] started worker PID=27233 for 'file5' (active=3)
[parent] worker PID=27220 finished (status 0, active=2)
  [worker PID=27232] done 'file4'
  [worker PID=27233] done 'file5'
[parent] worker PID=27232 finished (status 0, active=1)
[parent] worker PID=27233 finished (status 0, active=0)
[parent] all 5 tasks processed (concurrency limit=3)

```